CESIM TEACHING NOTE

Financial Analysis Microsimulation

Contents

1. Simulation Overview

The Financial Analysis Microsimulation is an immersive, experiential, hands-on simulation that equips participants with practical experience in investment analysis. Through the evaluation of financial statements, market conditions, and company-specific data, participants learn to construct optimized portfolios that reflect sound risk-return tradeoffs. This module develops core competencies in financial analysis, industry assessment, ratio interpretation, and strategic decision-making, providing a realistic and academically rigorous foundation for careers in finance and investment.

Learning Objectives

Upon completion of this simulation, participants will be able to:

- Develop proficiency in analyzing financial statements, including income statements and balance sheets
- Apply financial ratio analysis to evaluate company performance and investment potential
- Assess industry trends and their impact on investment decisions
- Formulate investment strategies based on quantitative and qualitative data
- Understand the interplay between financial metrics and market dynamics

Target Audience

- Graduate students in business, finance, or related fields seeking practical experience in financial analysis and investment decision-making
- MBA candidates and participants in executive education programs aiming to strengthen strategic finance skills
- Mid-level finance and business professionals looking to advance their analytical and portfolio management capabilities
- Senior executives and corporate leaders involved in financial planning, investment strategy, or capital allocation decisions

Time Required

- Approximately 45-60 minutes for individual completion of the simulation, with an additional 30-45 minutes recommended for debriefing and discussion.

Case Scenario and Winning Criteria

In this simulation, participants take on the role of financial analysts responsible for evaluating a set of companies within a defined sector, with particular focus on emerging trends and strategic opportunities. They are expected to conduct comprehensive financial analysis, interpret key performance indicators, and evaluate both qualitative and quantitative data to inform well-grounded investment decisions.Operating within a predefined investment budget, participants analyze financial statements, industry conditions, and company-specific data to construct an optimized investment portfolio. Investment decisions should align with broader strategic objectives—such as sustainability—while reflecting a strong grasp of core financial metrics, including liquidity, leverage, efficiency, profitability, and market valuation ratios.

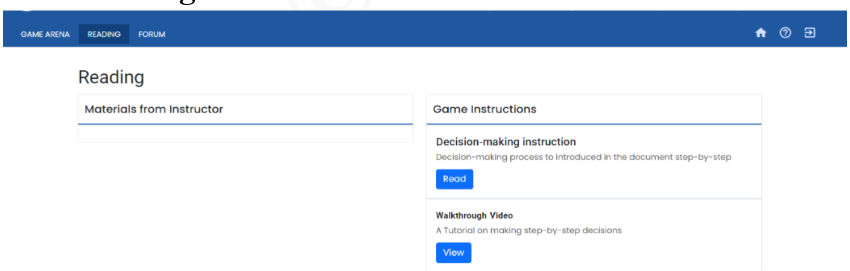
Success is measured by the effectiveness of the portfolio mix and the accuracy with which participants calculate and apply relevant financial ratios in support of their strategy.

Simulation Structure

- The microsimulation is organized into independent rounds, each presenting either consistent scenarios or allowing the instructor to customize market conditions to simulate decision-making challenges throughout the company's growth journey.

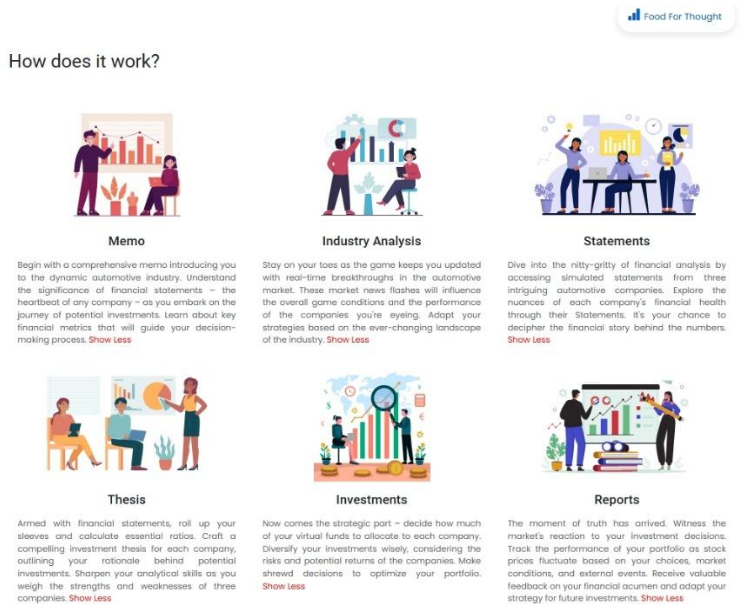
2. Pre-Simulation Preparation

a. **Review Background Materials**



- Login as participant to explore the platform
- Read the decision-making guide thoroughly
- Watch the walkthrough video for practical insights
- Understand the company context and business objectives

b. **Familiarize with the Interface**

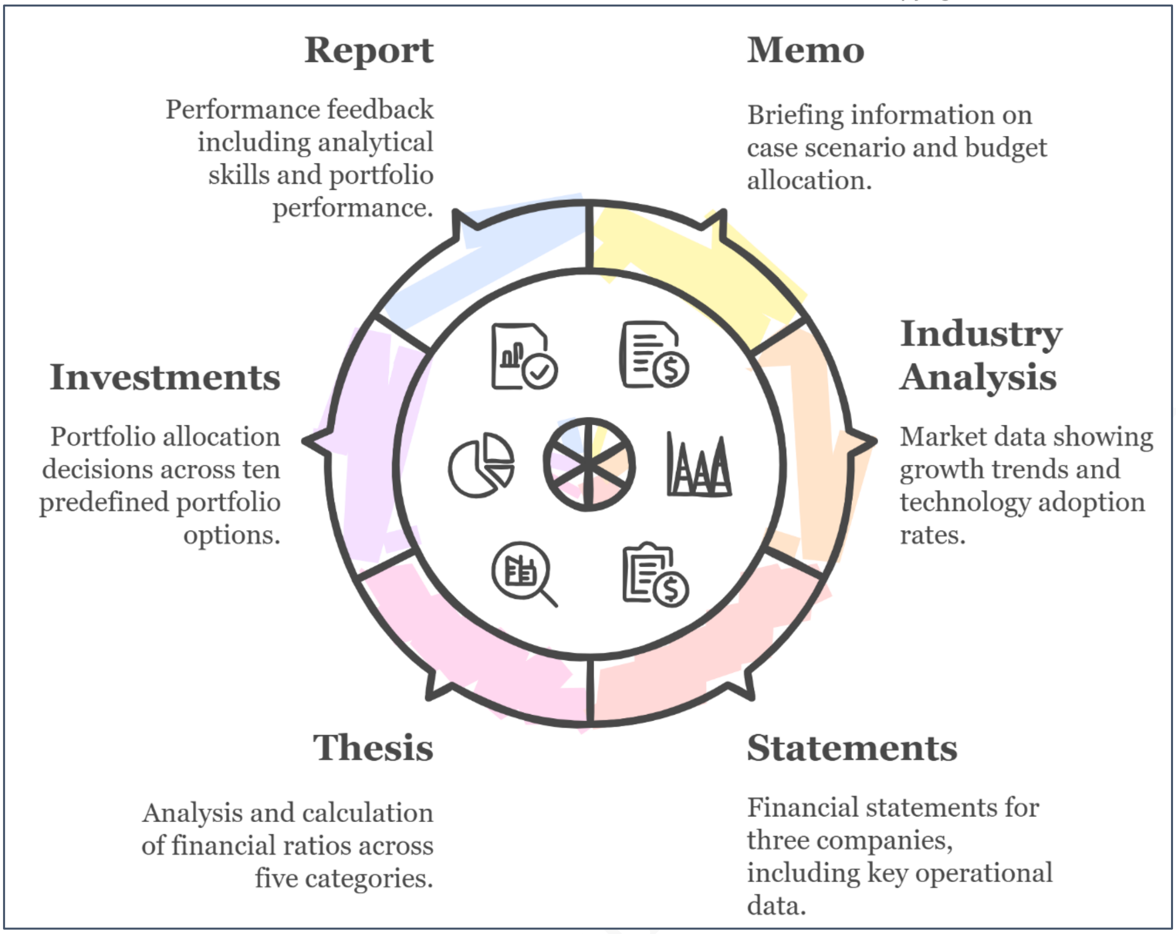


- Explore the Game Arena layout and understand the sequence of decision tabs
- **Memo**: Contains the briefing information and budget allocation.
- **Industry Analysis**: Provides market data showing growth trends and technology adoption rates across regions
- **Statements**: Contains financial statements for three companies including income statements, balance sheets, and key operational data
- **Thesis**: Where participants analyze and calculate financial ratios across five categories
- **Investments**: Where portfolio allocation decisions are made across ten predefined portfolio options
- **Report**: Performance feedback including radar chart of analytical skills and portfolio performance metrics

c. Understanding Decision-Making Process

Participants navigate through the following key decision areas:

- **Industry Analysis**: Examining market size projections and technology adoption trends across regions
- **Statements Analysis**: Reviewing income statements, balance sheets, and key operational data for the three client companies
- **Financial Ratio Calculation**: Computing and interpreting 15+ financial ratios across liquidity, leverage, efficiency, profitability, and market value categories
- **Investment Portfolio Construction**: Selecting from 10 predefined portfolio allocation options, ranging from balanced to more concentrated strategies



d. Core Algorithm

The simulation uses an algorithm that calculates investment returns based on:

- Historical financial performance of each company (profitability trends, growth rates)
- Market growth forecasts across various regions
- Technology adoption rates across different regions
- The participant's allocation of investment across the three companies
- Risk factors associated with each company's financial structure (debt levels, liquidity position)

e. Key Instructor-Only Parameters

Backend parameters include:

- Future market growth rates by region
- Technology adoption acceleration factors
- Company-specific growth multipliers
- Risk adjustment factors based on debt levels
- Market share data for each company
- Historical share price data and projected trends
- Portfolio performance calculations for each of the 10 predefined allocation options

f. Teaching with this Information

Use the backend parameters to:

- Explain why certain portfolio allocations yield higher returns
- Illustrate how regional market dynamics affect company performance
- Demonstrate the relationship between financial metrics and market performance
- Highlight the trade-offs between growth potential and financial stability

g. Common Participant Challenges

- Overreliance on a single financial metric without considering the broader financial picture
- Difficulty interpreting the significance of regional technology adoption patterns
- Struggle to connect industry trends with company-specific financial performance
- Confusion about the impact of debt structure on investment risk
- Overvaluing current revenue size versus profitability margins and growth potential

h. Instructor Pre-Implementation Checklist for Simulation Delivery

To ensure a smooth and impactful simulation experience for your learners, please review and complete the following preparatory steps:

- Familiarize Yourself with the Simulation: Complete a test run of the simulation to understand its structure, decision flow, and pacing. This allows you to better anticipate participant questions and challenges.
- Review Simulation Mechanics: Examine the instructor-only backend settings and key variables to understand how decisions influence outcomes.
- Align with Course Objectives: Identify the core concepts from your course that align with the simulation to reinforce key learning goals.
- Prepare Contextual Examples: Bring 2–3 real-world examples from relevant industries to illustrate successful and unsuccessful strategic decisions during the debrief.
- Review Instructor Insights and Scoring: Familiarize yourself with the instructor account, leaderboard and performance indicators available to instructors. These tools will help guide feedback and encourage reflective learning.
- Ensure Participant Access; Confirm that all participants have registered using the right link / code and are able to access the simulation platform before the session begins.
- Test Technical Readiness: Coordinate with your IT team (if applicable) to ensure all devices, software, and internet connectivity are ready and compatible with the simulation requirements.

3. Facilitation Roadmap

a. Pre-Simulation Briefing

1. Introduce the simulation context and objectives

   Case Summary: The simulation places participants in the role of an investment analyst at Dupont Venture. With a budget of 100,000 INR, participants must analyze three companies in the sustainable transportation sector, evaluate their financial statements, calculate key ratios, and make portfolio allocation decisions. The companies operate across various regions with different technology focuses, and participants must consider both current financial performance and future growth potential aligned with market trends.
2. Review the key financial concepts participants will apply
3. Explain how investment decisions impact returns
4. Clarify the role of industry analysis in financial decision-making
5. Outline the evaluation criteria for successful portfolio construction
6. Provide a brief overview of the simulation interface and navigation

b. During Simulation

1. Monitor participant progress through the different sections
2. Provide clarification on financial concepts as needed
3. Encourage participants to thoroughly analyze market share data by region and technology
4. Remind participants to consider both quantitative metrics and qualitative factors
5. Prompt deeper thinking about the connection between industry trends and financial performance
6. Guide participants who may be stuck on particular analyses

c. Debrief Structure

1. **Performance Review**: Discuss the range of portfolio returns achieved by participants
2. **Analysis Approaches**: Explore different analytical approaches used
3. **Key Insights**: Identify the most critical financial indicators that influenced successful decisions
4. **Industry Connections**: Discuss how industry trends shaped company performance
5. **Strategic Implications:** Consider broader strategic lessons for investment analysis
6. **Real-World Application**: Relate simulation insights to real-world investment decisions

4. Critical Decision Points Analysis

Decision Point 1: Industry Trend Analysis

Key Decision: How to interpret market size and technology adoption trends across regions
Optimal Approach:

- Analyze the market size data showing significant growth over the next decade
- Compare technology adoption rates across different regions
- Note the varying adoption rates of emerging technologies across different markets
- Recognize the emergence of newer technologies in specific regions
- Consider the implications of different regional growth rates

Common Pitfalls:

- Overlooking regional differences in technology adoption
- Failing to consider growth trajectories for different technologies
- Ignoring the implications of market size growth for company revenue potential
- Underestimating the importance of emerging technologies despite their smaller current market share.

Region	Technology Type 1 - Electric	Technology Type 2 Hybrid	Technology Type 3 Hydrogen	Technology Type 4 Combustion	Market Size	Growth Rate
Region 1	30%	20%	5%	45%	$65 billion	9.45%
Region 2	40%	17%	8%	35%	$419 billion	2.16%
Region 3	25%	15%	10%	50%	$36.1 billion	1.61%

Decision Point 2: Financial Statement Analysis

Key Decision: How to evaluate and compare the financial health of the three target companies

Optimal Approach:

- Conduct comprehensive analysis of income statements, focusing on revenue growth, cost structure, and profitability
- Examine balance sheet components, particularly asset composition and liability structure
- Compare key operational metrics such as capacity utilization and sales units
- Note distinctive characteristics:
    - Client 1 [Electra Motors]: Strong cash position, high profit margin, zero debt
    - Client 2 [Titan Motors]: Significant revenue growth, capacity expansion, low R&D investment
    - Client 3 [GreenSpeed Technologies]: Superior EBITDA margin, diversified product portfolio, significant debt

Common Pitfalls:

- Focusing solely on bottom-line profit without analyzing revenue sources and cost structures
- Overlooking cash position and debt levels when assessing financial stability
- Failing to connect operational metrics with financial outcomes
- Missing the significance of R&D investment disparities

Company	Revenue (M INR)	EBITDA Margin	Net Margin	Debt-to-Equity	Cash Position (M INR)	Sales Units	Capacity Units
Client 1 [Electra]	16,305	20.6%	17.1%	0%	6,486	4,665	4,820
Client 2 [Titan]	12,130	4.1%	4.7%	0%	4,238	3,743	2,060
Client 3 [GreenSpeed]	5,417	21.6%	16.4%	51%	4,020	1,490	4,200

Decision Point 3: Ratio Analysis and Valuation

Key Decision: How to calculate and interpret key financial ratios to assess investment potential

Optimal Approach:

- Calculate profitability ratios:
  - Return on Equity: Client 1 (30.0%), Client 2 (9.2%), Client 3 (24.5%)
  - Operating Margin: Client 1 (19.6%), Client 2 (3.7%), Client 3 (19.4%)
- Analyse efficiency ratios and asset utilization metrics.
- Evaluate market valuation metrics:
  - P/E Ratio: Client 1 (17.1), Client 2 (24.9), Client 3 (11.6)
  - Share Price: Client 1 (1,700 INR), Client 2 (315 INR), Client 3 (295 INR)
- Develop a comparative framework to rank companies on key financial dimensions

Common Pitfalls:

- Over-reliance on a single ratio category (e.g., focusing only on profitability)
- Failing to consider the relationship between different ratios
- Not accounting for industry context when interpreting ratio values
- Missing the significance of capacity utilization (Client 1: 97%, Client 2: 182%, Client 3: 35%)

Ratio Category	Key Metrics	Client 1 [Electra]	Client 2 [Titan]	Client 3 [GreenSpeed]
Liquidity	Current Ratio	17.4	14.2	30.4
Leverage	Debt-to-Equity	0.0	0.0	0.51
Efficiency	Asset Turnover	1.67	1.83	0.96
Profitability	ROE	30.0%	9.2%	24.5%
Market Value	P/E Ratio	17.1	24.9	11.6

Decision Point 4: Portfolio Allocation

Key Decision: How to allocate the 100,000 INR investment budget across the three companies

Optimal Approach:

- Balance the portfolio based on risk-return profiles of each company
- Consider both financial performance and alignment with industry trends
- Allocate higher percentages to companies with strong financial fundamentals and growth potential
- Ensure diversification across different product focuses and market segments
- Select an optimal portfolio mix from the provided options, weighing the unique strengths of each company

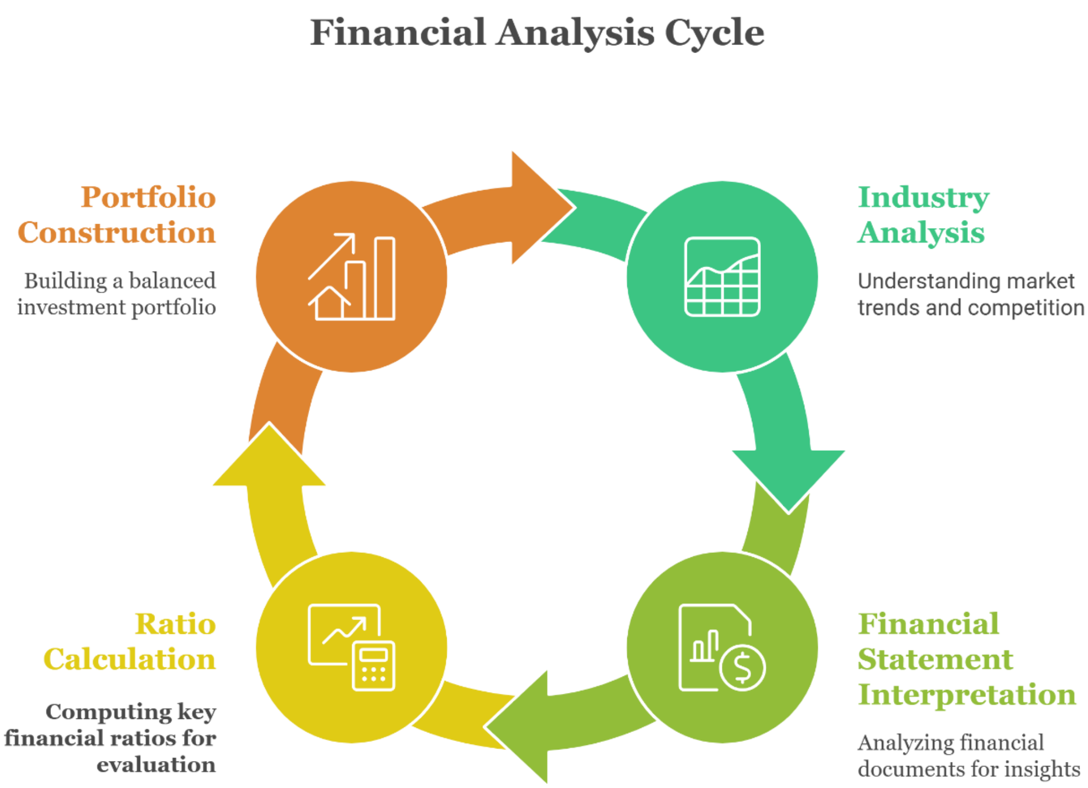
Common Pitfalls:

- Allocating funds based solely on past performance without considering future potential
- Creating an unbalanced portfolio overly concentrated in a single company
- Failing to connect industry trends with company-specific positioning
- Ignoring the risk factors associated with different financial structures.

Portfolio Option	Client 1 [Electra]	Client 2 [Titan]	Client 3 [GreenSpeed]	Rationale
Portfolio 7 (Optimal)	60%	20%	20%	Maximizes exposure to Client 1's strong financials while maintaining diversification
Portfolio 2	50%	30%	20%	Similar to optimal but with higher exposure to lower-margin Client 2
Portfolio 4	20%	20%	60%	Overweighted toward Client 3's margins but higher debt risk
Portfolio 1	33%	33%	33%	Equal allocation ignores significant company differences

5. Understanding Decision Linkages and System Dynamics

The Financial Analysis Microsimulation features interconnected decision areas that influence each other, creating a dynamic system where choices in one area impact options and outcomes in others.



This systems dynamic shows how industry trends provide context for interpreting financial statements, which in turn inform ratio analysis, leading to portfolio construction decisions. The feedback loop represents how investment performance reinforces understanding of which financial indicators were most relevant to success.

Summary of Core Decision Areas & Key Parameters

Decision Area	Key Parameters	Impact on Outcomes
Industry Analysis	Market growth rates, technology adoption trends by region, regional market sizes	Provides context for company potential and growth trajectories
Financial Statement Analysis	Revenue, EBITDA margins, debt levels, cash positions	Forms foundation for understanding company health and operations
Ratio Analysis	ROE, operating margins, P/E ratios	Enables comparative evaluation and investment prioritization
Portfolio Construction	10 predefined allocation options ranging from balanced to concentrated strategies	Directly determines investment returns and risk exposure

6. Performance Analysis Framework

Summary of Key Performance Indicators and Interpretation

KPI	Decision Area	Report Location	Interpretation
Portfolio Return	Investment Allocation	Results Report	Measures the financial performance of selected portfolio option
Analytical Skills Radar	Overall Performance	Radar Chart	Displays scores across rigor, synthesis, structuring, and business judgment dimensions
Financial Ratio Scores	Ratio Analysis	Scores Table	Shows performance across five ratio categories (liquidity, leverage, efficiency, profitability, market value)
Company Ownership	Investment Allocation	Investment Table	Indicates percentage ownership achieved in each company based on allocation and share prices
Percentile Ranking	Overall Performance	Leaderboard	Compares participant performance against peers and benchmarks

Mapping Decision Areas to Performance Metrics

The simulation evaluates participant performance across multiple dimensions:

1. **Financial Analysis Accuracy**: How well participants interpreted financial statements and calculated key ratios
2. **Industry Trend Integration**: How effectively industry trends were factored into investment decisions
3. **Portfolio Optimization**: How well the investment allocation balances risk and return
4. **Strategic Alignment**: How the portfolio positioning supports long-term industry trends
5. **Business Judgment**: Overall quality of decision-making based on available information.

7. Theoretical Connections by Simulation Section

Decision Section	Simulation Area	Theory	Summary
Industry Analysis	Market Dynamics	Porter's Five Forces, PESTEL Analysis	Understanding competitive forces and macro-environmental factors affecting the industry
Financial Statements	Income Statement, Balance Sheet	Financial Statement Analysis, Accounting Theory	Interpreting financial reports to assess operational performance and financial position of companies at different stages of technology adoption
Ratio Analysis	Financial Metrics	Ratio Analysis, Financial Performance Evaluation	Using standardized metrics to compare companies and evaluate financial health across multiple dimensions
Portfolio Construction	Investment Allocation	Modern Portfolio Theory, Asset Allocation	Balancing risk and return through strategic investment distribution, considering both financial metrics and industry positioning
Market Trends	Technology Adoption	Diffusion of Innovation, Technology Adoption Lifecycle	Understanding how new technologies gain market acceptance and grow at different rates across regions

Critical Teaching Points

1. **Integrated Analysis**: Successful financial analysis requires integration of industry context, company-specific data, and forward-looking perspectives.
2. **Beyond the Numbers**: Financial statements reveal strategic positioning within the technological landscape, particularly regarding investments in future capabilities.
3. **Comparative Context**: Financial ratios gain meaning when compared across companies with different strategic focuses and market positions.
4. **Risk-Return Balance**: Investment decisions must balance stability, growth potential, and innovation capabilities.
5. **Strategic Metrics**: Traditional financial metrics should be supplemented with assessments of innovation potential and alignment with industry transitions.

8. Sample Optimal Solution

Based on comprehensive analysis of the provided financial data and market trends, the optimal investment approach would be:

Ratio Analysis	Electra Motors	Titan Motors	GreenSpeed Technologies
Liquidity Ratio			
- Current Ratio	17.4	14.3	30.4
- Acid-test Ratio	14.8	11.1	28.3
- Cash Ratio	13.5	10.0	26.9
Leverage Ratio			
- Debt to Equity Ratio	0.0	0.0	0.5
- Debt Ratio	0.0	0.0	0.3
- Interest Coverage Ratio	NA	NA	NA
Efficiency Ratio			
- Asset Turnover Ratio	1.8	1.9	0.9
- Inventory Turnover Ratio	17.8	8.4	10.3
- Days Sales in Inventory Ratio	20.6	43.5	35.3
Profitability Ratio			
- Gross Margin Ratio	0.2	0.1	0.3
- Operating Margin Ratio	0.2	0.0	0.2
- Return on Asset Ratio	0.3	0.1	0.2
- Return on Equity Ratio	0.3	0.1	0.2
Market Value Ratio			
- Book value per share Ratio	331.4	137.1	103.5
- Earnings per share Ratio	99.3	12.7	25.4
- Price-earning Ratio	17.1	24.9	11.6

Portfolio Allocation

- **Portfolio 7**:
  - Client 1 [Electra Motors]: 60% (60,000 INR)
  - Client 2 [Titan Motors]: 20% (20,000 INR)
  - Client 3 [GreenSpeed Technologies]: 20% (20,000 INR)

Decision-Making Rationale

Client 1 [Electra Motors] (60%)

- Strongest financial position with substantial cash reserves and zero debt
- Highest profitability with 17.1% net margin and 30% ROE
- Operating in key emerging technology segments
- Strong unit sales growth (17.7% increase)
- Reasonable P/E ratio of 17.1 compared to market
- Share price of 1,700 INR indicates market confidence
- High capacity utilization (97%) suggesting operational efficiency

Client 2 [Titan Motors] (20%)

- Impressive revenue growth (23.6% increase)
- Expanding manufacturing capacity (33% increase)
- Transitioning to focus on emerging technologies
- Zero debt position provides financial stability
- Lower allocation due to weaker profitability metrics (4.7% net margin)
- High capacity utilization (182%) indicating potential production constraints
- Lower share price (315 INR) offers potential upside

Client 3 [GreenSpeed Technologies] (20%)

- Superior EBITDA margin (21.6%)
- Diversified technology portfolio across all segments
- Expanded regional presence to include additional markets
- Lower allocation due to higher debt levels (51% debt-to-equity ratio)
- Concerning decrease in revenue (-6.3%)
- Low capacity utilization (35%) suggesting operational inefficiencies
- Attractive P/E ratio of 11.6 indicates potential undervaluation

This portfolio balances strong current financial performance (Client 1) with growth potential (Client 2) and technological diversification (Client 3), while maintaining alignment with industry shifts toward emerging technologies. The allocation acknowledges regional trends including differential adoption rates across markets.

9. Common Pitfalls and Teaching Interventions

Pitfall 1: Overemphasizing Bottom-Line Profits

Symptoms: Participants focus exclusively on net income without considering revenue trends, margin stability, or operational efficiency.

Intervention: Guide participants to examine profit margins rather than absolute profit figures. For example, note that while Client 1 has the highest profit, Client 3 achieves a similar net margin with only one-third the revenue.

Pitfall 2: Neglecting Balance Sheet Analysis
Symptoms: Analysis concentrates on income statement metrics while ignoring asset composition, debt levels, and liquidity.
Intervention: Emphasize the importance of Client 3's significant long-term debt (51% debt-to-equity ratio) compared to the zero-debt positions of Clients 1 and 2, particularly in a capital-intensive industry facing technological disruption.

Pitfall 3: Ignoring Industry Context
Symptoms: Financial analysis occurs in isolation from industry trends and technological shifts. **Intervention:** Ask targeted questions about how regional differences in technology adoption might impact each company's prospects. For example, how might Client 1 benefit from Region 2's higher adoption rate of emerging technologies?

Pitfall 4: Unbalanced Portfolio Construction
Symptoms: Investment allocation heavily concentrated in a single company based on one standout metric.
Intervention: Present Portfolio 4 (20% Client 1, 20% Client 2, 60% Client 3) as a cautionary example that overweight a company with significant debt despite attractive margins, creating unnecessary risk concentration.

Pitfall 5: Static Analysis
Symptoms: Focusing only on current period figures without considering trends and growth rates.
Intervention: Direct participants to calculate year-over-year changes, such as Client 2's 25% increase in sales units compared to Client 3's 20% increase and Client 1's 18% increase.

10. Instructor Tips and Insights

Suggested Discussion Starters

1. "How did you balance the tension between current financial performance and future growth potential in your investment decisions?"
2. "What significance did you attach to the varying debt structures of the three companies, and how did this impact your portfolio allocation?"
3. "Which financial ratios proved most valuable in your comparative analysis, and why did you prioritize these particular metrics?"
4. "How did you interpret the capacity utilization rates of the three companies, and what implications did you draw about their operational efficiency?"
5. "To what extent did regional market differences influence your assessment of each company's strategic positioning?"
6. "What risks did you identify in each company's financial structure, and how did you factor these into your investment strategy?"

7. "How might changing technology adoption rates across regions affect the long-term prospects of companies with different technological focuses?"
8. "What conclusions did you draw from the R&D spending patterns of the three companies, and how did this affect your view of their innovation potential?"
9. "How did you weigh the trade-off between high margins and high growth when evaluating these investment opportunities?"
10. "What additional information would have been valuable to refine your investment decision, and how would you have used it?"

FAQs

General Microsimulation Questions

Q: How is the performance score calculated?

A: The performance score is based on the accuracy with which participants calculated and applied key financial ratios, including Liquidity, Leverage, Efficiency, Profitability, and Market Value ratios. The total score reflects overall performance across these measures.

Q: Can I revisit previous sections to adjust my analysis?

A: Yes, you can navigate back to previous sections until you submit your final decisions at the Decision Checklist.

Strategy Questions

Q: Should I prioritize companies with higher current profitability or those with better growth prospects?

A: This represents a classic investment tradeoff. Consider that Client 1 offers high current profitability while Client 2 shows stronger revenue growth. The optimal approach typically balances both considerations.

Q: How important is debt in evaluating these companies?

A: In capital-intensive industries, debt structure is crucial. Client 3's significant debt (51% debt-to-equity ratio) represents a risk factor compared to the zero-debt positions of Clients 1 and 2.

Conceptual Questions

Q: How does technological disruption affect financial analysis approaches?

A: Technological disruption introduces greater uncertainty in forecasting and requires analysts to consider non-traditional metrics such as innovation capacity (R&D spending), technology adoption rates, and strategic positioning alongside traditional financial measures

Q: How might environmental and social considerations affect investment decisions?

A: These factors are increasingly important in investment analysis. Companies with stronger environmental and social credentials may command premium valuations and potentially benefit from regulatory advantages and changing consumer preferences.

Conclusion

The **Financial Analysis Microsimulation** immerses participants in applying advanced financial analysis skills within the context of an industry undergoing rapid technological transformation. Through this experience, learners gain practical insights into interpreting financial statements, calculating and evaluating key ratios, and making well-reasoned strategic investment decisions.

Key decision-making moments require participants to balance quantitative financial metrics with qualitative industry insights, such as regional adoption patterns of emerging technologies. Success depends not only on evaluating a company's current financial health but also on assessing its positioning for future market opportunities, informed by projected growth trends and shifts in technology adoption across different regions.

The most successful participants demonstrate the ability to synthesize diverse data points into a coherent investment strategy, weighing:

- Financial strength and stability
- Growth potential across markets and technologies
- Diversification and adaptability in the face of technological disruption

These analytical skills are directly transferable to real-world finance and investment, where professionals must navigate complex financial landscapes while accounting for broader market dynamics and ongoing technological change.

Supplementary Information: Formulas

Liquidity Ratio	**Formulas**
- Current Ratio	(Inventory + Recievables + Cash)/(Revolver debt + Payables)
- Acid-test Ratio	(Recievables + Cash)/(Revolver debt + Payables)
- Cash Ratio	(Cash)/(Revolver debt + Payables)
Leverage Ratio	
- Debt to Equity Ratio	(Long-term+Short-term debt)/(Total Equity)
- Debt Ratio	(Long-term+Short-term debt)/(Total Asset)
- Interest Coverage Ratio	(Operating Profit-EBIT)/(Net Financing Expense), if NFE negative NA
Efficiency Ratio	
- Asset Turnover Ratio	(Sales Revenue)/(Average Total Assets)
- Inventory Turnover Ratio	(Variable production cost + features cost + JV manufacturing + transportation cost)/(Average Inventory)
- Days Sales in Inventory Ratio	365/Inventory Turnover Ratio
Profitability Ratio	
- Gross Margin Ratio	(Sales - Variable production cost - features cost - JV manufacturing - transportation cost)/(Sales)
- Operating Margin Ratio	(Operating Profit-EBIT)/(Revenue)
- Return on Asset Ratio	(Net Profit)/(Total Assets)
- Return on Equity Ratio	(Net Profit)/(Total Equity)
Market Value Ratio	
- Book value per share Ratio	(Total Equity)/(Number of Shares Outstanding)
- Earnings per share Ratio	(Net Profit)/(Number of Shares Outstanding)
- Price-earning Ratio	(Share Price, INR)/(Earnings per share Ratio)